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L21_L22 (2).pdf
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A lecture scribe is a detailed record of the lecture for exam reference which includes detailed definitions and annotations, assumptions and conditions, statement of theorems or results, proof or proof sketches, step by step derivations and worked examples only if they appear in the given content. No new content should be introduced, only the given content(the uploaded ppt here) should be used to generate the scribe.

The scribe should the follow the given format-
Header information and topic name
Introduction and overview which includes brief outlines of all the topics covered Main body content which should include all the points mentioned above required for a scribe
Important notes/ remarks
Conclusion

Here is your **lecture scribe** strictly based on the uploaded PPT (Lectures 21 & 22), with **no added content**, following the required format.

Lecture Scribe

Course: CSE400 – Fundamentals of Probability in Computing
Lectures: 21 & 22
Topic: Inequalities and Tail Bounds
Instructor: Dhaval Patel, PhD
Dates: March 24 and 26, 2025 L21_L22 (2)

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1. Introduction and Overview

These lectures introduce the concept of **tail probabilities** and methods to bound them using inequalities. The main focus is on understanding rare events and deriving bounds

when exact computation is difficult.

Topics Covered:

- Definition and motivation of **tails**
- Difficulty in computing tail probabilities
- **Markov's Inequality**
  - Definition, proof idea, and examples
- **Chebyshev's Inequality**
  - Definition, proof idea, and examples
- Introduction to **Chernoff Bounds** (key idea only)
- Running examples using **Binomial and Poisson distributions**
- Real-life applications and interpretations

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2. Main Body Content

2.1 Tails and Motivation

Definition

The tail of a random variable X is:

$$P(X > x)$$

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

Key Idea

- Mean → average behavior (easy)
- Tail → rare extreme events (hard)

Why are tails hard to compute?

- Requires summing many small probabilities L21_L22 (2)

2.2 Tail Examples

Example 1: Load Distribution

- Distribute n jobs among n servers randomly

- Question: Probability a server gets $\geq k$ jobs

Model:

$$X \sim \text{Binomial}(n, p)$$

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Interpretation:

- Most servers get few jobs
- Occasionally, one server gets many jobs → tail event

Example 2: Poisson Arrivals

- Jobs arrive at rate λ jobs/hour

Model:

$$X \sim \text{Poisson}(\lambda)$$

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Interpretation:

- Most hours have moderate arrivals
- Sometimes bursts occur → tail event L21_L22 (2)

2.3 Why Tail Bounds?

Problem

Exact computation:

- Binomial and Poisson tails involve **long sums**
- Computationally complex

Idea

Instead of exact value:

- Find an **upper bound**
- Easier to compute
- Still provides a guarantee

Key Idea:

Approximate safely instead of computing exactly L21_L22 (2)

2.4 Running Example

- Flip a fair coin n times
- $X =$ number of heads

$$X \sim \text{Binomial}(n, \tfrac{1}{2})$$

$$E[X] = \frac{n}{2}$$

Question:

$$P(X \geq \tfrac{3n}{4})$$

- Tail = unusually large number of heads

2.5 Markov's Inequality

Key Idea

- Uses only the mean
- Large values are rare

Statement

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof (Idea)

- Consider only values $X \geq a$
- Replace X by a

$$E[X] = \sum x \cdot p_X(x) \geq \sum_{x \geq a} x \cdot p_X(x) \geq \sum_{x \geq a} a \cdot p_X(x) = a \cdot P(X \geq a)$$

$$\Rightarrow P(X \geq a) \leq \frac{E[X]}{a}$$

Running Example (Markov)

$$P\left(X \geq \frac{3n}{4}\right) \leq \frac{E[X]}{3n/4} = \frac{n/2}{3n/4} = \frac{2}{3}$$

Remark:

- Bound is **constant** (does not depend on n)
- Clearly a **poor bound**

Real-Life Example (Income)

- Average income = 10,000

$$P(X \geq 50,000) \leq \frac{10,000}{50,000} = 0.2$$

Conclusion:

- At most 20% people can earn 50,000

Solved Example 1 (Exam Grades)

- $E[X] = 70$, threshold $a = 90$

$$P(X \geq 90) \leq \frac{70}{90} = \frac{7}{9} \approx 0.778$$

Result:

At most 77.8% students receive an A

Solved Example 2 (Server Load)

- $E[X] = 5000$, $a = 20000$

$$P(X \geq 20000) \leq \frac{5000}{20000} = \frac{1}{4} = 0.25$$

Result:

At most 25% chance of overload

2.6 Chebyshev's Inequality

Key Idea

- Uses mean + variance

- Controls deviation from mean

Statement

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Proof Idea

Apply Markov to $(X-\mu)^2$:

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2} = \frac{\mathrm{Var}(X)}{a^2}$$

Running Example (Chebyshev)

- $X \sim \mathrm{Binomial}(n, \frac{1}{2})$
- $\mu = \frac{n}{2}, \quad \mathrm{Var}(X) = \frac{n}{4}$

$$P\left(X \geq \frac{3n}{4}\right) \leq P\left(|X - \tfrac{n}{2}| \geq \tfrac{n}{4}\right) \leq \frac{n/4}{(n/4)^2} = \frac{4}{n}$$

Key Insight:

- Bound decreases as n increases
- Better than Markov

Real-Life Example (Exam Scores)

- Mean = 70, Std dev = 10

$$P(|X-70| \geq 30) \leq \frac{10^2}{30^2} = \frac{1}{9}$$

Result:
At most $\frac{1}{9}$ students are far from mean

Solved Example 1 (Grades with Variance)

- $\mu = 70, \sigma = 5 \Rightarrow \sigma^2 = 25$
- $a = 20$

$$P(X \geq 90) \leq P(|X-70| \geq 20) \leq \frac{25}{400} = \frac{1}{16} = 0.0625$$

Result:

Much tighter than Markov bound (0.778)

Solved Example 2 (Network Latency)

- $\mu = 50, \sigma^2 = 16$
- Range: [30, 70] ⇒ $a = 20$

$$P(|X - 50| \geq 20) \leq \frac{16}{400} = \frac{1}{25} = 0.04$$

Result:

At most 4% packets fall outside range

2.7 Chernoff Bound (Key Idea Only)

Progression of Ideas

- Markov: bound using X
- Chebyshev: bound using $(X - \mu)^2$
- Chernoff: bound using e^{tX}

Key Insight

- Exponential grows very fast
- Large values become much larger
- Tail events get amplified

Conclusion:

Better transformation ⇒ tighter tail bounds L21_L22 (2)

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3. Important Notes / Remarks

- Tail probabilities represent **rare events**
- Exact computation is often **complex due to long summations**
- Markov inequality:
  - Simple but **loose bound**
  - Uses only mean
- Chebyshev inequality:
  - Uses variance
  - Gives **tighter bounds than Markov**
- Bounds improve with more information (mean → variance → exponential)

- Markov bounds only **right tail**
- Chebyshev bounds **deviation from mean (both tails)**

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4. Conclusion

These lectures establish the foundation for analyzing **rare events in probability** using inequalities. Since exact tail probabilities are difficult to compute, **tail bounds provide practical and computable guarantees**. The progression from **Markov to Chebyshev to Chernoff** demonstrates how incorporating more information and stronger transformations leads to **tighter and more useful bounds**, especially in real-world systems such as server loads, network latency, and data center performance.

If you want, I can also convert this into **clean LaTeX notes or exam-ready short notes**.

Sources